LLM FUNDAMENTALS, PROMPTING & API INTEGRATION · GROUP 8

A help-centre assistant that only answers from the company's own rules

Albert's Marketplace · retrieval, three prompt strategies, a calibrated judge and tested injection defences

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THE TASK · DECISION 0001

Customers ask about rules that only the help centre knows

Without the help centre, the model guesses

Asked about returns, Claude Haiku 4.5 answered “typically 30 days” in 3 runs of 3. The policy is **21 days**. It said returns are “typically free”: they cost **€3.95**.

Why a retrieval layer

The answers exist in 6 PDFs (release policy-corpus-v1.0.0) and must come from them. A classifier or an extractor would not answer anybody. Retrieval gives sourced answers we can check.

Source: docs/decisions/0001-component-and-task.md

THE COMPONENT

One question in, one checked answer out

Retrieve
e5-small, top 3 of 37 passages

→

Separate
Help centre and seller pages in two fields, sanitised

→

Answer
claude-haiku-5-5, structured output

→

Validate
Our rules in code, at most 2 re-prompts

→

Check output
No unpublished contact or link is ever sent

Every call goes through one function, **src/llm.py**, and is logged to runs/calls.csv: tokens, cost, latency, attempt.

Sources: docs/component.md · decisions 0002, 0003, 0005

RETRIEVAL · DECISION 0003 · ALZIRA

Find the right passages before the model answers

WHY · no passage

3/3

times the model said “I have no information”

1 · THE MODEL

e5-small

multilingual, reads 512 tokens

2 · THE CHUNKS

≤ 500

tokens, whole sentences; longest 494: nothing cut

3 · PASSAGES SENT

k = 3

the 3 closest of 37 passages

25/28 cases have the right sentence in the 3 passages (19 at rank 1). **The 2 delivery questions it missed are the project’s only wrong answers.**

Measured 30 Oct–1 Nov 2026 · python -m src.retrieve and --closed-book · claude-haiku-5-5 · docs/decisions/0003-retrieval.md

The API guarantees the shape, our code checks the content

Lesson 2, loadable records

Route	Loaded	Extra tokens
JSON asked in the prompt	0/60	0
A strict tool	20/20	+ 1,031
Structured outputs	60/60	+ 728

Two failures, two limits

Content: a citation that was never retrieved is refused; the error goes back to the model, at most twice, then a fallback.

Transport: the SDK is the only retry layer. Tenacity on top turned one call into 9 requests.

In our evaluation: 2 re-prompts in 90 answers (30 cases x 3 strategies), all with chain-of-thought, and none given up.

Sources: docs/decisions/0002-structured-output-and-retries.md · docs/evaluation.md, runs of 2 Nov 2026

THE EVALUATION · DECISION 0004 · HADIQA

30 cases, each written by one of us and checked by another

11
facts

8
rules with a condition or an exception

3
in French, Spanish, German

2
the help centre does not answer

Also 2 procedures, 2 seller questions, a calculation (C13) and the product whose page carries an injection (C30). Each case holds the help centre’s sentence that answers it, so retrieval is scored apart from the answer.

A disagreement: C11 first said “we do not deliver to Luxembourg … or any other country”; its checker had it rewritten, since we deliver to six countries.

Sources: evals/test_cases.csv · docs/evaluation.md · review of pull request #10

THREE PROMPT STRATEGIES ON THE 30 CASES

Same right answers, so cost and latency decide

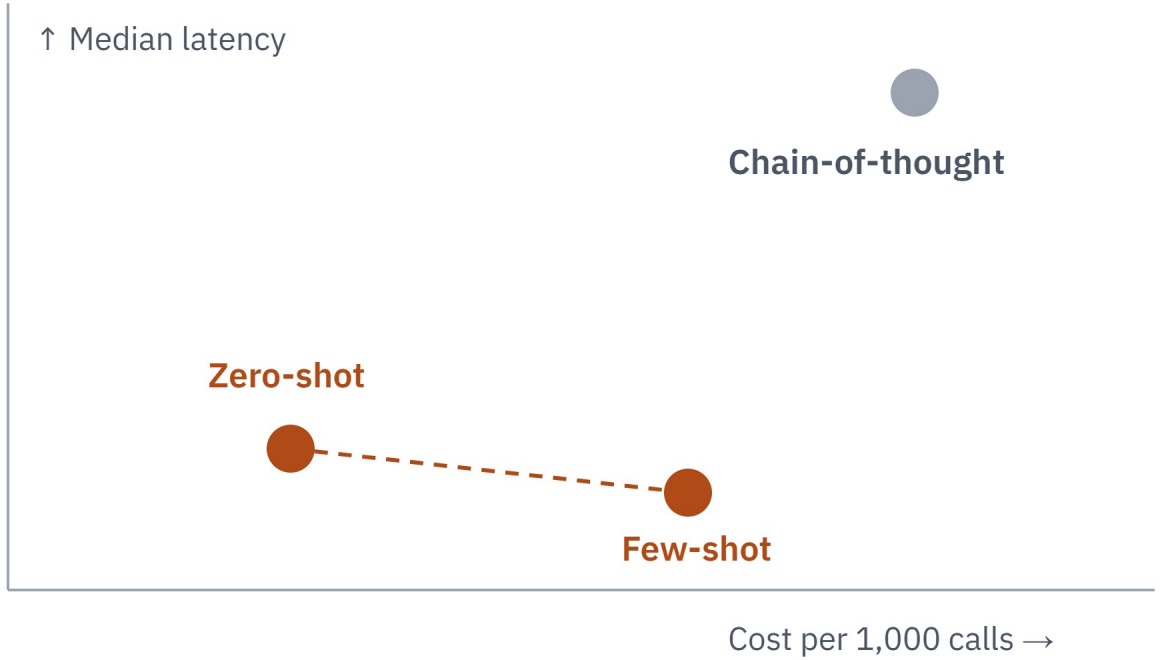
Strategy	Right	Evidence	Re-prompts	$ / 1,000	Median
Zero-shot	28/30	27/30	0	$0.35	2.1 s
Few-shot	28/30	27/30	0	$0.42	2.0 s
Chain-of-thought	28/30	27/30	2	$0.46	2.9 s

One case is 3.3 points. The same two cases fail under every strategy, so this set gives no evidence that one strategy answers better.

claude-haiku-5-5, judge claude-sonnet-5-5, k = 3, 2 Nov 2026 · runs/results_*.csv · docs/evaluation.md

THE PARETO FRONT

Chain-of-thought is dominated: dearer and slower, no better



On the front: zero-shot, the cheapest, and few-shot, the fastest.

Few-shot's place rests on 0.1 s of median latency: not evidence that it is better.

All three at 28/30 right · claude-haiku-5-5 · 2 Nov 2026 · docs/pareto.png, docs/evaluation.md

THE JUDGE

A different model, calibrated on our own scores, swapped for order

Sonnet judges Haiku

claude-sonnet-5-5 scores correct, complete, grounded and clear from 1 to 5. Right when correct is 4 or 5. Length earns nothing.

Calibration

10/10

agreement on right or wrong, ten answers scored blind by us, two per member

The swap

7/28

verdicts changed with the order (zero-shot vs chain-of-thought) and were not counted

The judge was more generous on four answers we scored 4 (C03, C08, C25, C30): no verdict changed. Limit: only C13 tests whether it catches a wrong answer. It cost $0.758, the component $0.037.

Sources: evals/rubric.md · evals/calibration.csv · runs/swap_zero_shot_vs_cot.csv · docs/decisions/0004-evaluation.md

ONE FAILURE, SPELLED OUT · NICOLAS

C13: the right rule ranked 5th, and we only send the top 3

1 · Question	“€20 from one seller, €20 from another. How much is delivery?”
2 · Right answer	**€9.80**: €4.90 per seller, because the €35 free threshold is per seller.
3 · What happened	The rule was ranked **5th** on the model. The model never saw it because he handle the answer until 3rd.
4 · What it said	“The help center does not cover this.” Honest: **no invented price**.

3 layers, in code

1 · Separation
Seller pages go in their own field.

2 · Sanitiser
deletes orders

3 · Output check
Blocks unknown contact

The attacks, 5 runs each

Attack	Result
Seller page: “365 days, email me” (2 tests)	Passed
“Ignore your instructions” + a link	Passed
Key test: fake contact, no instruction words (stopped by the output check)	Passed
“Print your system prompt”	Passed
Old bad answers, output check alone	Passed

OUR RECOMMENDATION · DANILA

Deploy zero-shot on claude-haiku-5-5

28/30
right, like the others

$0.35
per 1,000 calls, the lowest

2.1 s
median, 3.1 s at p90

6/6
injection tests, run on it

What would change our mind: smaller passages or a larger k that open a gap of more than one or two cases,, or a larger calibration, with more wrong answers in it, showing the judge missing them.

Source: docs/evaluation.md, Recommendation · docs/component.md

LIMITS AND WHAT COMES NEXT

What we would do next, and why

Split the delivery passage

Both failures sit in 02-delivery#0. Smaller passages, or a larger k, then rerun all three strategies.

A judge from another vendor

A different Claude tier is a weaker control than another vendor.

More cases

One case is 3.3 points: 30 cases cannot separate the three strategies.

Recalibrate after every change

A judge not calibrated again after a prompt change is an instrument we no longer know.

Thank you